



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Chapter 1

Part 3

Fundamental and Elements of Logic

Why Are We Studying Logic?

Some of the reasons:

- Logic is the foundation for computer operation
- Logical conditions are common in programs

Example:

Selection: `if (score <= max) { ... }`

Iteration: `while (i < limit && list[i] != sentinel) ...`

- All manner of structures in computing have properties that need to be proven (and proofs that need to be understood).

Example: Trees, Graphs, Recursive Algorithms, ...

- Programs can be proven correct.
- Computational linguistics must represent and reason about human language, and language represents thought (and thus also logic).

PROPOSITION

A **statement** or a **proposition**, is a declarative sentence that is **either TRUE or FALSE, but not both.**

Example:

- 4 is less than 3.
- 7 is an even integer.
- Washington, DC, is the capital of United State.

Example

- i) Why do we study mathematics?
- ii) Study logic.
- iii) What is your name?
- iv) Quiet, please.

} False

The above sentences are not propositions. Why ?

(i) & (iii) : is question, not a statement.
(ii) & (iv) : is a command.

Example

- i) The temperature on the surface of the planet Venus is 800 F.
- ii) The sun will come out tomorrow.

Propositions? Why?

- Is a statement since it is either true or false, but not both.
- However, we do not know at this time to determine whether it is true or false.

CONJUNCTIONS

- Compound propositions formed in English with the word **“and”**
- Formed in logic with the caret symbol (**“ $\wedge$ ”**)
- **True** only when **both participating propositions are true.**

TRUTH TABLE: This tables aid in the evaluation of **compound propositions.**

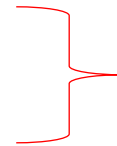
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

True (T), False (F)

Example

p : 2 is an even integer

q : 3 is an odd number



propositions

$p \wedge q$



symbols

: 2 is an even integer and 3 is an odd number



statements

p : today is Monday

q : it is hot

$p \wedge q$: today is Monday and it is hot

Example

Proposition

p : 2 divides 4

q : 2 divides 6

Symbol & Statement

$p \wedge q$: 2 divides 4 and 2 divides 6.

or,

$p \wedge q$: 2 divides both 4 and 6.

Example

Proposition

p : 5 is an integer

q : 5 is not an odd integer

Symbol & Statement

$p \wedge q$: 5 is an integer and 5 is not an odd integer.

or,

$p \wedge q$: 5 is an integer but 5 is not an odd integer.

DISJUNCTION

- Compound propositions formed in English with the word “**or**”,
- Formed in logic with the caret symbol (“ **$\vee$** ”)
- **True** when **one or both** participating propositions are true.

The **truth table** for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example

p : 2 is an integer

q : 3 is greater than 5

$p \vee q$: 2 is an integer or 3 is greater than 5

p : $1+1=3$

q : A decade is 10 years

$p \vee q$: $1+1=3$ or a decade is 10 years

Example

p : 3 is an even integer

q : 3 is an odd integer

$p \vee q$

3 is an even integer or 3 is an odd integer
or

3 is an even integer or an odd integer

NEGATION

Negating a proposition simply flips its value. Symbols representing negation include:

$\neg x$, $\bar{x}$, $\sim x$, x' (NOT)

Let p be a proposition.
The negation of p , written $\neg p$ is the statement obtained by negating statement p .

The **truth table** of $\neg p$

p	$\neg p$
T	F
F	T

Example

p : 2 is positive

$\neg p$: 2 is not positive.

p : 4 is less than 3

$\neg p$: 4 is not less than 3.

CONDITIONAL PROPOSITIONS

Let *p* and *q* be propositions.

*“if **p**, then **q**”*

is a statement called a **conditional proposition**,
written as

$$p \rightarrow q$$

CONDITIONAL PROPOSITIONS

The **truth table** of $p \rightarrow q$
 $\Rightarrow$ *Cause and effect relationship*

FALSE if
 $p = \text{True}$
 and q
 $= \text{false}$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

TRUE if
 both
 true or
 $p = \text{false}$
 for any
 value of
 q

Example

p : today is Sunday

q : I will go for a walk

p* → *q : If today is Sunday, then I will go for a walk.

p : I get a bonus

q : I will buy a new car

p* → *q: If I get a bonus, then I will buy a new car

Example

p : $x/2$ is an integer.

q : x is an even integer.

$p \rightarrow q$: if $x/2$ is an integer, then x is an even integer.

BICONDITIONAL

Let p and q be propositions.

“ p if and only if q ”

is a statement called a **biconditional proposition**,
written as

$$p \leftrightarrow q$$

BICONDITIONAL

The truth table of $p \leftrightarrow q$:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

p : my program will compile

q : it has no syntax error.

$p \leftrightarrow q$: My program will compile if and only if it has no syntax error.

p : x is divisible by 3

q : x is divisible by 9

$p \leftrightarrow q$: x is divisible by 3 if and only if x is divisible by 9.

- The compound propositions **Q** and **R** are made up of the propositions $p_1, \dots, p_n$.
- **Q** and **R** are logically equivalent and write,
$$Q \equiv R$$
provided that given any truth values of $p_1, \dots, p_n$, either **Q** and **R** are **both true** or **Q** and **R** are **both false**.

Example

$$Q = p \rightarrow q \quad R = \neg q \rightarrow \neg p$$

Show that, $Q \equiv R$

The **truth table** shows that, $Q \equiv R$

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Example

Show that, $\neg (p \rightarrow q) \equiv p \wedge \neg q$

The **truth table** shows that, $\neg (p \rightarrow q) \equiv p \wedge \neg q$

p	q	$\neg(p \rightarrow q)$ proposition it, then	$p \wedge \neg q$ and
T	T	F	F
T	F	T	T
F	T	F	F
F	F	F	F

PRECEDENCE OF LOGICAL CONNECTIVES

Precedence of logical connectives is as follows:

not	$\neg$	↑	Highest
and	$\wedge$		
or	$\vee$		
If...then	$\rightarrow$		
If and only if	$\leftrightarrow$		Lowest

Example

Construct the truth table for, $A = \neg(p \vee q) \rightarrow (q \wedge p)$
if, then

Solution

p	q	$(p \vee q)$	$\neg(p \vee q)$	$(q \wedge p)$	A
T	T	T	F	T	T
T	F	T	F	F	T
F	T	T	F	F	T
F	F	F	T	F	F

$p \rightarrow q$

Logic and set theory go very well together. The previous definitions can be made very succinct:

$x \notin A$ if and only if $\neg(x \in A)$

$A \subseteq B$ if and only if $(x \in A \rightarrow x \in B)$ is True

$x \in (A \cap B)$ if and only if $(x \in A \wedge x \in B)$

$x \in (A \cup B)$ if and only if $(x \in A \vee x \in B)$

$x \in A - B$ if and only if $(x \in A \wedge x \notin B)$

$x \in A \Delta B$ if and only if $(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$

symmetric
difference

$A \oplus B$

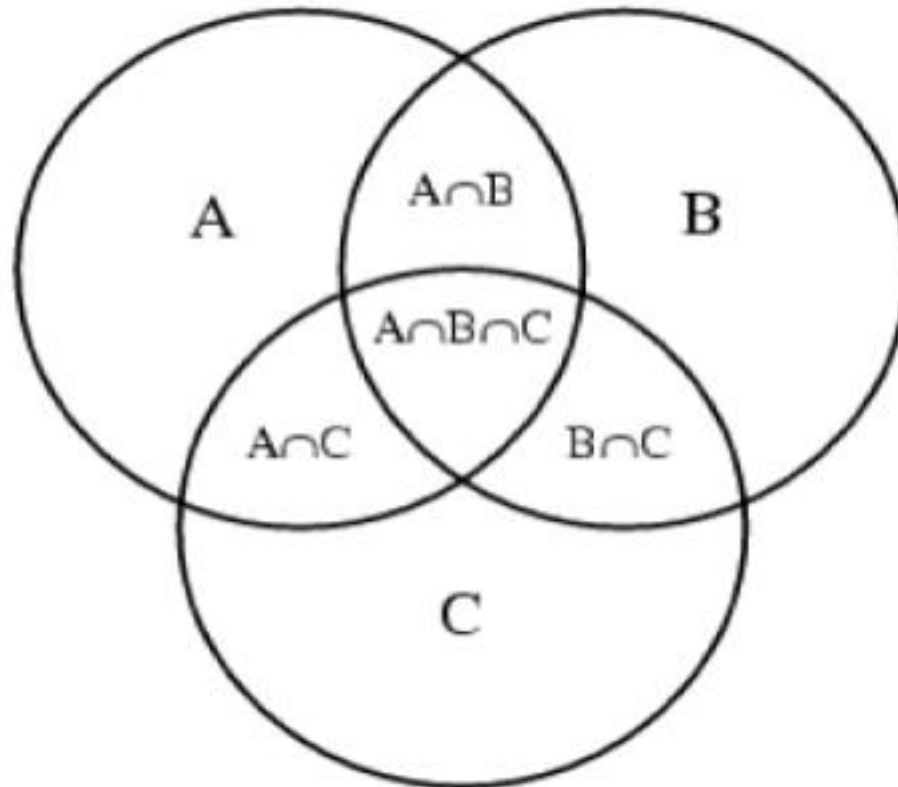


$x \in A'$ if and only if $\neg(x \in A)$

$X \in P(A)$ if and only if $X \subseteq A$

Venn Diagrams

Venn Diagrams are used to depict the various unions, subsets, complements, intersections etc. of sets.



Logic and Sets are closely related

Tautology

$$p \vee q \leftrightarrow q \vee p$$

$$p \wedge q \leftrightarrow q \wedge p$$

$$p \vee (q \vee r) \leftrightarrow (p \vee q) \vee r$$

$$p \wedge (q \wedge r) \leftrightarrow (p \wedge q) \wedge r$$

$$p \vee (q \wedge r) \leftrightarrow (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \leftrightarrow (p \wedge q) \vee (p \wedge r)$$

$$p \wedge \neg q \leftrightarrow p \wedge \neg(p \wedge q)$$

$$p \wedge \neg(q \vee r) \leftrightarrow (p \wedge \neg q) \wedge (p \wedge \neg r)$$

$$p \wedge \neg(q \wedge r) \leftrightarrow (p \wedge \neg q) \vee (p \wedge \neg r)$$

$$p \wedge (q \wedge \neg r) \leftrightarrow (p \wedge q) \wedge \neg(p \wedge \neg r)$$

$$p \vee (q \wedge \neg r) \leftrightarrow (p \vee q) \wedge \neg(r \wedge \neg p)$$

$$p \wedge \neg \vee (q \wedge \neg r) \leftrightarrow (p \wedge \neg q) \vee (p \wedge r)$$

Set Operation Identity

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A - B = A - (A \cap B)$$

$$A - (B \cap C) = (A - B) \cup (A - C)$$

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$A \cap (B - C) = (A \cap B) - (A \cap C)$$

$$A \cup (B - C) = (A \cup B) - (C - A)$$

$$A - (B - C) = (A - B) \cup (A \cap C)$$

The above identities serve as the basis for an "algebra of sets".

Logic and Sets are closely related

Tautology

$$p \wedge p \leftrightarrow p$$

$$p \vee p \leftrightarrow p$$

$$p \wedge \neg(q \wedge \neg q) \leftrightarrow p$$

$$p \vee \neg(q \wedge \neg q) \leftrightarrow p$$

Contradiction

$$p \wedge \neg p$$

$$p \wedge (q \wedge \neg q)$$

$$p \wedge \neg p$$

Set Operation Identity

$$A \cap A = A$$

$$A \cup A = A$$

$$A - \emptyset = A$$

$$A \cup \emptyset = A$$

Set Operation Identity

$$A - A = \emptyset$$

$$A \cap \emptyset = \emptyset$$

$$A - A = \emptyset$$

The above identities serve as the basis for an "algebra of sets".

Theorem for Logic

Let p , q and r be propositions.

Idempotent laws:

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

Truth table

p	$p \wedge p$	$p \vee p$
T	T	T
F	F	F

Theorem for Logic

Double negation law:

$$\neg \neg p \equiv p$$

Commutative laws:

$$p \wedge q \equiv q \wedge p$$

$$p \vee q \equiv q \vee p$$

Associative laws:

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

Theorem for Logic

Distributive laws:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

p	q	r	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

PROVE

Theorem for Logic

Absorption laws:

$$p \wedge (p \vee q) \equiv p$$

$$p \vee (p \wedge q) \equiv p$$

PROVE

p	q	$p \wedge (p \vee q)$	$p \vee (p \wedge q)$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	F	F

Theorem for Logic

De Morgan's laws:

$$\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$$

$$\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$$

The **truth table** for $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$

p	q	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F
T	F	F	F
F	T	F	F
F	F	T	T

Thank You